

Unit 1: Introduction (6 Hrs.)

E-Governance — An Overview

E-Government

Electronic government (or e-Government) is application of Information and Communication Technologies (ICTs) to government functions and procedures with the purpose of increasing efficiency, transparency and citizen participation.

e-Government is embedded in combinations of political conditions as well as cultural, technological and organizational changes designed to support and drive a profound transformation in government units.

E-Governance

E-Governance is dealing with all regulations and policies to control the services provided by the e-Government. However, the E-Government is an electronic government, which should be regulated by the E-Governance.

Moreover, e-Governance refers to the utilization of information and communication technology (ICT) for providing government services, disseminating information, communication operations with the general public.

Elements of E-Governance:

Basic elements of e-governance are:

1. Government
2. Citizens
3. Investors/Businesses

Types of E-Governance:

E-governance is of 4 types:

1. Government-to-Citizen (G2C):

The Government-to-citizen mentions the government services that are acquired by the familiar people. Most of the government services come under G2C. Similarly, the primary aim of Government-to-citizen is to supply facilities to the citizens. It also helps the ordinary people to minimize the time and cost to carry out a transaction. A citizen can retrieve

the facilities anytime from anywhere. Similarly, spending the administrative fee online is also possible due to G2C. The facility of Government-to-Citizen allows the ordinary citizen to outclass time limitation. It also focuses on geographic land barriers.

2. Government-to-business (G2B):

The Government-to-business is the interchange of services between Government and Business firms. It is productive for both government and business firms. G2B provides access to pertinent forms needed to observe. It also contains many services interchanged between business sectors and government.

Similarly, the Government-to-business provides timely business information. A business organization can have easy and easy online access to government agencies. G2B plays a important role in business development. It upgrades the efficiency and quality of communication and transparency of government projects.

3. Government-to-Government (G2G):

The Government-to-Government mentions the interaction between different government departments, firms and agencies. This increases the efficiency of government processes. In G2G, government agencies can share the same database using online communication. The government departments can work together. This service can increase international discretion and relations.

G2G services can be at the local level or at the international level. It can convey with both global government and local government. It also provides safe and secure inter-relationship between domestic or foreign government. G2G builds a universal database for all members to upgrade service.

4. Government-to-Employee (G2E):

The Government-to-Employee is the internal part of G2G section. It aims to bring employees together and improvise knowledge sharing. It provides online facilities to the employees. Similarly, applying for leave, reviewing salary payment record and checking the balance of holiday. The G2E sector yields human resource training and development. So, G2E is also the correlation between employees and government institutions.

Advantages of E-Governance:

1. **Speed:** Technology makes communication swifter. Internet, smartphones have enabled instant transmission of high volumes of data all over the world.
2. **Saving Costs:** A lot of Government expenditure goes towards the cost of buying stationery for official purposes. Letters and written records consume a lot of stationery. However, replacing them with smartphones and the internet can save crores of money in expenses every year.
3. **Transparency:** The use of e-governance helps make all functions of the business transparent. All Governmental information can be uploaded onto the internet. The citizens access specifically access whichever information they want, whenever they want it, at their convenience.

However, for this to work the Government has to ensure that all data as to be made public and uploaded to the Government information forums on the internet.

4. **Accountability:** Transparency directly links to accountability. Once the functions of the government are available, we can hold them accountable for their actions.

Disadvantages of E-Governance:

1. **Loss of Interpersonal Communication:** The main disadvantage of e-governance is the loss of interpersonal communication. Interpersonal communication is an aspect of communication that many people consider vital.
2. **High Setup Cost and Technical Difficulties:** Technology has its disadvantages as well. Specifically, the setup cost is very high and the machines have to be regularly maintained. Often, computers and

internet can also break down and put a dent in governmental work and services.

3. **Illiteracy:** A large number of people are tech-illiterate and do not know how to operate computers and smartphones. E-governance is very difficult for them to access and understand.
4. **Cybercrime/Leakage of Personal Information:** There is always the risk of private data of citizens stored in government servers being stolen. Cybercrime is a serious issue, a breach of data can make the public lose confidence in the Government's ability to govern the people.

The key difference between e-Government & e-Governance

- e-Government is a system whereas e-Governance is a functionality.
- e-Government means the application of ICT in government operations, as a tool to make a better government. e-Governance, on the other hand, implies the use of ICT in transforming and supporting functions and structures of the system.
- e-Government is a one-way communication protocol. On the contrary, e-Governance is a two-way communication protocol. (government to citizen and vice versa)
- e-Governance is the part of e-Government. e-Governance never comes alone.

Benefits of E-Government

1. E-Government enhances citizen participation in governance policy making.
2. It increases transparency and accountability of development processes in national institutions.
3. It enhances access to information and knowledge resources to all areas.
4. It improves the delivery of basic services to the overall population.

5. Provides citizen satisfaction and improve the efficiency of administration.
6. E- Government provides a corruption-free governance.
7. It greatly simplifies the process of information accumulation for citizens and businesses.
8. It empowers people to gather information regarding any department of government and get involved in the process of decision making.
9. E-Government strengthens the very fabric of democracy by ensuring greater citizen participation at all levels of governance.
10. E-Government leads to automation of services, ensuring that information regarding every work of public welfare is easily available to all citizens, eliminating corruption

Issues in E-Governance Applications and the Digital Divide

The digital divide is the gap between those with Internet access and those without it. But the digital divide is multifaceted and includes many factors such as access, affordability, quality, and relevance. As Michael Kende wrote, "the digital divide is not a binary." Here are some of the things that lead to disparities in Internet access:

- **Availability:** Is there available access to the Internet in your area? Is there a nearby point of connection to the Internet? If yes, this is just the first step to having Internet access.
- **Affordability:** Is that access affordable? How does the cost compare to other essential goods? What percentage of your income do you need to pay for access?
- **Quality of service:** Are the upload and download speeds sufficient for the local needs of Internet users?
- **Relevance:** Does the connected community have the necessary skills and technologies? Is there local interest and understanding of the relevance of Internet access? Are there locally available mobile apps? Is there content in the local language and relevant to the people in the community?
- **Additional divides:** Other areas that can create digital inequality include security, interconnectivity, digital literacy, and access to equipment.

Some of the vivid effects of the digital divide are elaborated below

Impact of the digital divide on education

The internet offers a rich reservoir of information and knowledge. Skills and expertise are well organized and conveyed over the internet making the use of computers to be a ubiquitous activity in the developed world. The access and availability of ICT have been associated with academic success and robust research activities since users can quickly make references. Education is a very dynamic sector and keeping up to date is crucial to success, the presence of internet access will ensure you get the latest trends and revolutionize your research skills. The inadequacy of ICT equipment has made the already weak education system in the developing countries even more ineffective.

Governments are now striving to avail ICT services to schools in an effort to improve the education standards. Installation of computer laboratories, provision of laptops to every student among other strategies have shown remarkable success rates.

Impact of the digital divide on the economy

ICT services is crucial for economic growth, and there exists a strong linear correlation between the two. Interestingly, socio-economic status is one of the major causes of the digital divide and it also a consequence of the digital divide. Penetration of internet enables people to engage in economically productive activities such as trade without much hassle. They can shop online and compare prices to get the best bargains, participate in online auctions and safely and securely transact online this translates to economic empowerment. For the sector that lacks access to technology, trade is made hectic, and returns get lowered this maintain the economic gap between the developed and the underdeveloped nations extensive. The rich countries get, and the emerging countries cannot compete with them hence remaining to depend on grants and donations.

Integration of ICT affects the success of a business and thus influence the circulation of resources among citizens.

Impact of the digital divide on social

The inclusion of ICT services has been crucial in enhancing social interactions among people. Social websites such as Facebook and Instagram have been useful in facilitating communication and maintaining touch with friend and relatives of close ties. Technology is very influential on the levels of relations among members of society since new social opportunities arise when we include ICT in our relationships. People can now keep in touch with friends or even make new friends even amidst tight schedule courtesy of chat rooms. Besides, internet harbors fertile information about peoples' culture and religion that are vital in guiding social relations.

Internet access offers person access to a broader range of opportunities thus creating a social divide between those who are enriched and those who are deficient. Such segregation creates a potential for social conflicts in the communities where the wealthy can have computers and access the internet while the poor are barred. Efforts to bridge this gap initiated at a personal level through unacceptable activities such as theft has led to the disruption of harmony in the society.

Impact of the digital divide on the society

The digital divide has contributed significantly to stratification in the community whereby there arises a class of persons with access to the internet and another class unable to use the ICT services either due to affordability issues or literacy levels. Some other factors that contribute to the digital divide in the society include age, race, and ethnicity. The digital divide has led to a rise of new alignments in the community whereby people are classified depending on ability to access internet services this is coupled with associated benefits, and therefore those with limited access to technology continue to lag behind in development matters.

Impact of the digital divide on culture

Technology affects different cultures in varied ways either on the positive or the negative. The integral components culture like relationships, communications, and art have been dramatically evolved by technology. However, it seems like some specific customs and cultural traditions have remained intact. The developed countries experience tremendous changes in their culture as they incorporate new cultures acquired from the internet while the developing countries with people having limited access to the internet have their culture unchanged. Because they do not have access to

technology, people in technology-poor nations are left behind. However, in order to narrow the digital divide, people need access to technology. To eliminate the ICT global divide is essential for the growth of developing countries. The development of information infrastructure is a quick way to economic growth for less developed nations. In as much as digital divide is essential in preserving certain cultures.

Evolution, Scope and Content of E-Governance

- Initiatives were taken up as early as 1972 by Chile
- Prof. Stafford Beer implemented for President Allende of Chile, the first governance software that would help the government survive a severe crisis.
 - Major contribution by US Vice President Al Gore in early 1990s which rooted worldwide in the information superhighway.
- Widespread in US, UK, Canada, Australia and India
- focus largely on development of infrastructure such as fiber optic networks.
- concept of Information Society or Knowledge Society evolved
- E-governance came into a formalized and focused manner with partial success to implement Information System in the government departments and public organizations
- During 1980s and 90s, govt. all over the world lagged behind the commercial world in accepting and implementing ICT.
 - Commercial and industrial world went far ahead all over the world in harnessing the potential of ICT in their core and peripheral activities. They used ICT to reach out to their customers and business partners, thereby impressively enhancing their quality, speed and convenience.
- Visible success cases of use of ICT – ATM services – 24 hour call center – E-Shopping
- Software export increase (banking, financial, aviation, industrial sector from India, Ireland, Israel, China)
- 1990s and 2000 - Development of ITES (IT enabled Society) – resulted in remote services like call centers, data entry
- Point to be noted –

All the above mentioned services are not e-Governance service

- Government last in the queue of institutions providing IT services
- Initial Efforts in E-governance:
 - Partial automation of existing paper based manual process
 - Did not result in significant Business Reengineering Process compared to private sector
 - No big changes seen in government enterprises
- May be because of Conservatism to change
- Rigidity of legislative body

- Major issues that has become highly relevant for large scale implementation of ICT in governance
 - Issue of Security
 - Privacy
 - Vulnerability (exposure) of public ICT infrastructure to crime
 - Potential to abuse, terrorism and general crime
 - Problems in social cohesion (unity)
 - Social Exclusion --- digital divide

- The scope of ICT implementation in government machinery result in
 - Improvement of efficiency and effectiveness of the executive function of government, including delivery of public services
 - Greater transparency of government to citizens and business permitting greater access to the information generated or collated by the government
 - Fundamental changes and improvement in relations between citizen and the state thereby improving the democratic process
 - Better interactions and relationships amongst different
 - Wings of the same government
 - State or local government within a country
 - Countries whose governance are web-enabled
 - Any e-governance activity/project involves appropriate

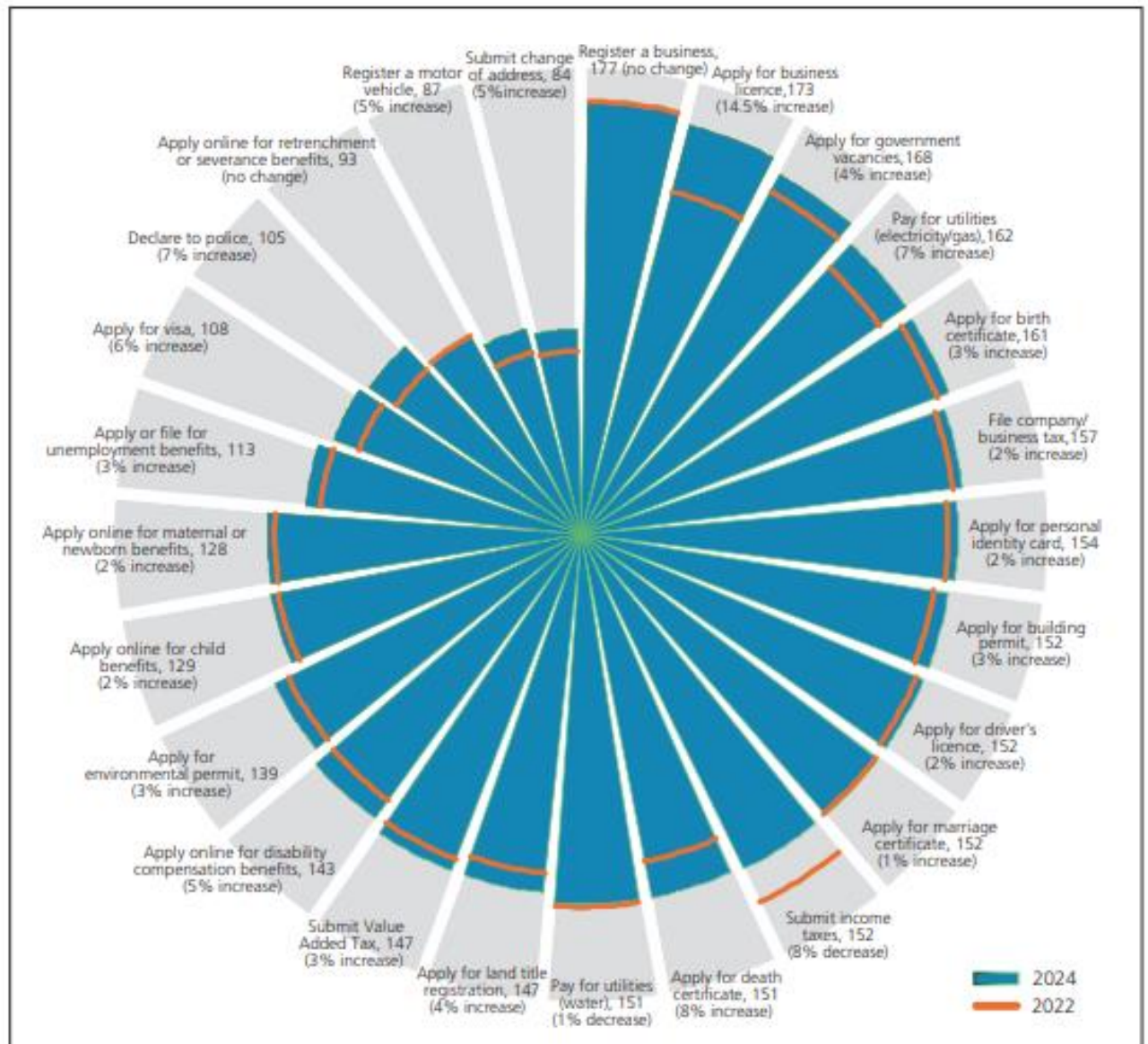
- Hardware and corresponding system software
- Networking of the hardware identified above- both the Internet and Intranet environment
- Application software along with appropriate database management software.

Present Global Trends of Growth In E-Governance

- According to press reports during the end of 2002 – Indicate a trend of global growth in e-governance utilization – Adults using internet world wide increase by 15% according to the findings of Second Government Online Study published by Taylor and Nelson. – 3 out of 10 citizens (30%) access government services online compared with only a quarter (26%) in 2001
- Government online services mostly used for – Search Information (24%) – Download Information (11%) – Increase in % to search info – 20% to 24% from Sep 2001 to Sep 2002
- Globally, online government transactions increased from just 6% to 7% during 2001 and 2002 and the percentage of those providing personal details to government increased from 7% to 8%
- According to UN E-government Survey report 2024 , 25 online services have been assessed, up from 22 for the last Survey cycle. The number of countries offering at least one of these online services has not changed, remaining at 189 (98 per cent) of the 193 Member States.

The most prevalent online transactional services remain the registration of a new business (177 countries) and applying for a business license (173 countries). The next most frequently offered online services include applying for government vacancies, paying utility bills (electricity and gas), applying for a birth certificate, and filing company taxes. The electronic submission of business taxes is offered by more countries than the online submission of income taxes, which is a departure from 2022. Tax-filing services are offered more frequently to businesses (157 countries) than to individuals (152 countries for income tax and 147 countries for Value Added Tax, or VAT). Among the least offered online services are changing an address (84 countries) and registering a motor vehicle (87 countries), though both services are offered by 5 per cent more countries in 2024 than in 2022.

Table 1.1 Trends in online transactional services, 2022-2024



Sources: 2022 and 2024 United Nations E-Government Surveys.

- E governance usage in different countries (between 2001- 2002) – Australia – significant increase from 31% to 46% – Turkey – 3% to 13% – Netherlands – 32% 41% – US – 34% to 43% – Japan – decrement by 4% - 17% to 13%
- The highest rates of growth have been in the provision of online services for older persons and migrants (a 14 per cent increase for each group) and for women (an increase of 11 per cent). Where specific services are in place, Governments provide information to older persons on retirement housing facilities and on how to apply for long-term care programs and choose options to receive care and support at home. Young people receive

information and support relating to specific employment programs, scholarships and government funding, and people living in poverty are provided with information on applying for government support.

- In 2020, around 80-90 per cent of Member States provide information to the public on sector specific policies and programs. Many countries also share government expenditures and budgets with the public, often in open formats. Compared with 2018, there has been an increase of approximately 50 per cent in the number of (Online Government Data) OGD portals that provide sector-specific information in machine-readable formats. The highest increase has been in the environment sector, where the number of countries with OGD portals offering machine-readable content has risen from 58 to 101 (74 per cent) since 2018

The number of United Nations Member States in the very high EGD (E-Government Development Index) group (with values ranging from 0.75 to 1.00) increased from 40 to 57, representing a 43 per cent increase between 2018 and 2020.

Table 1.2. Country EGD groupings and rating classes within EGD groups (from highest to lowest) in 2020 and movement between EGD groups between 2018 and 2020

Very High EGD 2020 (Greater than 0.75)	Rating Class	Movement within group	High EGD 2020 (Between 0.50 and 0.75)	Rating Class	Movement within the group	Middle EGD 2020 (Between 0.25 to 0.50)	Rating Class	Movement within the group	Low EGD 2020 (Less than 0.25)	Rating Class	Movement within the group
Denmark	VH	none	Serbia	HV	none	Lebanon	MH	H to M	Guinea-Bissau	LM	None
Republic of Korea	VH	none	Albania	HV	none	Eswatini	MH	none	Democratic People's Republic of Korea	LM	None
Estonia	VH	none	Brunei Darussalam	HV	none	Guyana	MH	none	Niger	L3	None
Finland	VH	none	Mexico	HV	none	Rwanda	MH	none	Chad	L3	None
Australia	VH	none	Barbados	HV	none	Syrian Arab Republic	MH	none	Central African Republic	L2	None
Sweden	VH	none	Mauritius	HV	none	Nepal	MH	none	Somalia	L2	None
United Kingdom of Great Britain and Northern Ireland	VH	none	Monaco	HV	VH to H	Tajikistan	MH	none	Eritrea	L1	None
New Zealand	VH	none	Georgia	HV	none	Timor-Leste	MH	none	South Sudan	L1	None
United States of America	VH	none	Qatar	HV	none	Lesotho	MH	none			
Netherlands	VH	none	Colombia	HV	none	Belize	MH	none			
Singapore	VH	none	Armenia	HV	none	Uganda	MH	none			
Iceland	VH	none	Ukraine	HV	none	Honduras	MH	none			
Norway	VH	none	Azerbaijan	HV	none	Côte d'Ivoire	MH	none			
Japan	VH	none	Peru	HV	none	Cuba	MH	none			

2020 UN E-GOVERNMENT SURVEY

- In 2001, security issue was the main concern which improved globally in 2002.
- 2001 – 14% felt secured with credit card and bank account numbers
- 2002 – this increased to 23%
- Highest levels of safety was seen in the Scandinavian Market (Denmark, Finland, Norway and Sweden) together with some South East Asian markets (Singapore and Hong Kong)--- around 1/3 of users felt safe.
- In contrast- 90% of Japanese felt accessing online services unsafe.
- Germany 82% and France 76%
- According to UN, (2011), Despite the primary web portals of the Governments, there are multiple channels of delivering E-Government Services and some of the major modes includes, - Smart phone internet access(feature-rich devices with maximum communication capabilities), Voice prompted interface (Multi-Lingual Capability), SMS mobile phone applications (Cost efficient),Social media applications (Provides two way interactions, free of cost).

Latest technological developments are a promising aspect for both public sector as well as ICTs industry across countries as discusses briefly in the table 1.3 below:

Table 1.3 Faster Technologies in the domain of E-Governance (Source: UN, 2018).

S.No.	Technology	Description / Application
1	Data, intelligent apps and analytics	Involves Automated Apps connecting Public institutions and citizens in areas like ticketing, documenting, etc. that are also enabled by digital payment facilities, analytics and report generation features.

2	Artificial Intelligence and Robotic Process Automation	Applies and simulates the processing capability of human brain towards machine learning, decision making and problem solving
3	Cyber-physical integration, edge computing and Intelligent "things"	Being evolved from Internet of Things (IoT), it attempts to integrate AI Technologies, Edge Computing and Sensor based data for effective monitoring and analysis.
4	Augmented Reality (AR) and Virtual Reality (VR)	Enables to Visualize scenarios, improve accuracy of decision making, etc.
5	High Performance and Quantum Computing	Understanding Complexity, and Detecting anomalies in data gathered helps in public sector domains like medicine, economic forecasting, traffic flows, meteorology, and tax collection. Quantum computing enables computing different results simultaneously and increases computing power exponentially.
6	Blockchain and Distributed Ledger Technologies	Improves the ways of storing information in a distributed manner

E-Governance Applications

The three variables of age, education, and broadband access at home significantly influence citizens' adoption of the Government Gateway (Dwivedi & Williams, 2008). Internal environment within the government and external environment—social, economic, cultural, technical, etc. play a useful role in the matter of e-government applications in the delivery of public services. Transparency and accountability in governance can be enhanced through e-government. The process and application of e-government projects are digitalizing government information. Applications of e-government include intra-governmental systems, online delivery of government services, online taxation, web based registration and licensing, and e-procurement.

While designing and developing e-government projects, application of e-government leads to:

1. Connections within government and between various government agencies/bodies;
2. Connections between government and citizens transforming public service delivery;
3. Connections between government and citizens strengthening transparency, responsiveness and accountability;
4. Connections within communities building social, economic and cultural development;
5. Connections between government and private sectors;
6. Connections between government and NGO supporting pre-determined goals and objectives; and
7. Connections between government and civil society organizations (CSOs) helping in delivery of public service.

Issues in E-Governance Applications:

1. Lack of strategies and Financial plan.
2. The project implementation is generally vendor driven.
3. Lack of standardization(For example, similar type of projects are carried by different state agencies using incompatible file formats and application standards)
4. Reverse compatibility of application with legacy systems are missing in several projects.
5. The IT infrastructures are obtained just before building the application or digitizing the data.
6. Physical security is emphasized whereas the logical application security is left to vendors in many cases.

7. Lack of understanding by the departments for the components of e-governance applications, which can be outsourced or can be carried out in-house.

E-Government Initiatives in Nepal

1972: Introduction of computer for census (IBM1401)

1974: Establishment of Electronic Data Processing Center

1982: First Private Overseas Investment in software development by establishing company for export, Data Systems International (p) LTD

1985: Distribution of Personal Computers

1990: Liberalization on imports of equipment

1992: Establishment of Computer Association of Nepal

1996: Establishment of the Ministry of Science & Technology

2000: Announcement of the first IT policy, "IT Policy 2000"

2001: Establishment of National Information Technology Center

2003: Establishment of High Level Commission for Information Technology

2007: Enactment of Electronics Transaction Act